

❖ Activity 3.2: Modular Architecture Setup

`/recommendations-details`: Returns detailed AI-generated product recommendations based on user budget, preferences, and selected category (Home, Party, or Jewelry).

```
# Step 6: Add an endpoint to get details for a specific recommendation
@app.get("/recommendation-details/{recommendation_id}")
async def get_recommendation_details(
    recommendation_id: str,
    request: Request,
    current_user: UserInDB = Depends(get_current_active_user)
):
    """Get the full details of a specific recommendation"""
    if current_user.username not in user_recommendations:
        raise HTTPException(status_code=404, detail="No recommendations found")

    # Find the recommendation with the given ID
    for item in user_recommendations[current_user.username]:
        if item.id == recommendation_id:
            return {
                "id": item.id,
                "timestamp": item.timestamp,
                "type": item.recommendation_type,
                "input": item.input_summary,
                "full_result": item.full_result
            }

    raise HTTPException(status_code=404, detail="Recommendation not found")
```

❖ Activity 3.3: CORS & Static Routing

`/history`: Retrieves the user's past recommendation queries and results for review or re-use.

```
# Step 7: Add a new endpoint to serve the history page
@app.get("/history", response_class=HTMLResponse)
async def history_page(request: Request, current_user: UserInDB = Depends(get_current_active_user)):
    """History page to view past recommendations"""
    return templates.TemplateResponse("history.html", {"request": request, "user": current_user})
```

❖ Activity 3.4: Startup and main function

`/startup`: Initializes essential application services and loads configuration settings when the FastAPI server starts.

```
@app.on_event("startup")
async def setup_session_cleanup():
    """Background task to clean up expired sessions"""
    async def cleanup_expired_sessions():
        while True:
            current_time = datetime.utcnow()
            # Check for sessions that have been inactive for more than 30 minutes
            expired_sessions = [
                username for username, session in active_sessions.items()
                if (current_time - session.last_activity).total_seconds() > 1800 # 30 minutes
            ]

            # Remove expired sessions
            for username in expired_sessions:
                if username in active_sessions:
                    print(f"Removing expired session for {username}")
                    del active_sessions[username]

            # Wait for 5 minutes before checking again
            await asyncio.sleep(300)

    # Start the background task
    asyncio.create_task(cleanup_expired_sessions())
```

`__main__`: Entry point of the FastAPI application that starts the server using `uvicorn` when the script is run directly.

```
# Main entry point
if __name__ == "__main__":
    import uvicorn
    print("Starting PocketSmart: AI Budget Planner...")
    uvicorn.run(app, host="0.0.0.0", port=8000)
```

MILESTONE 4: UI Development :

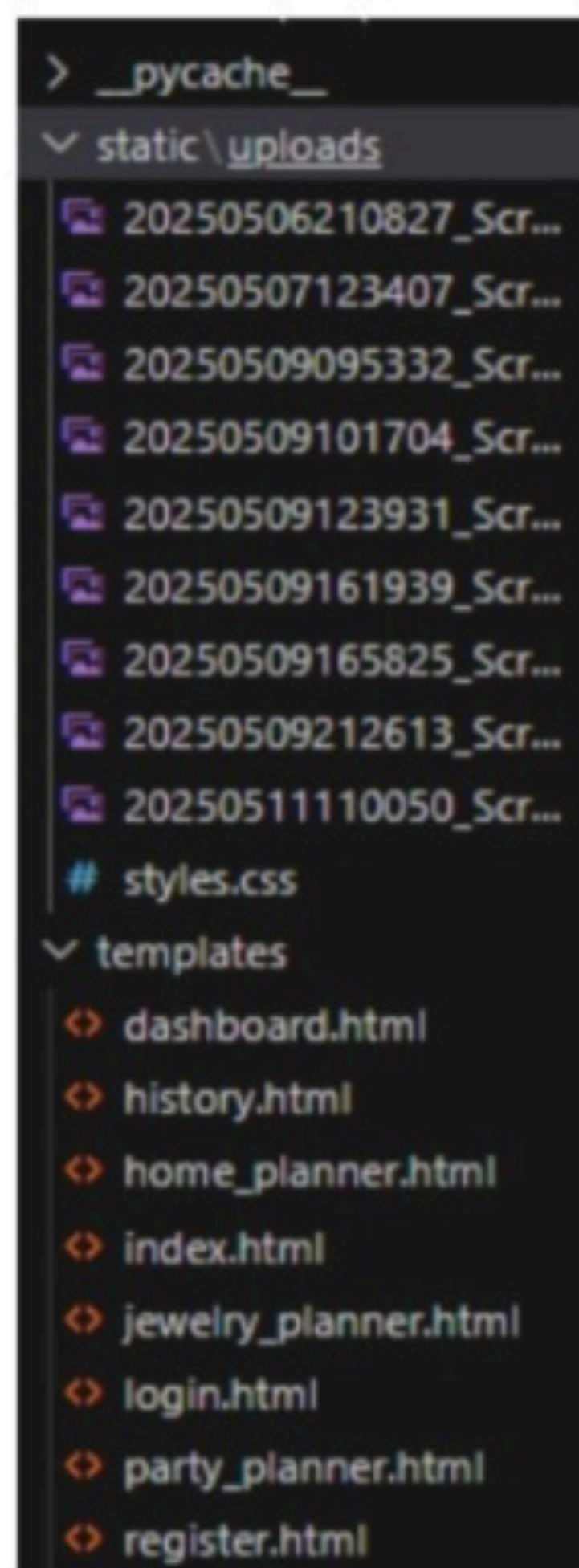
- **Core Goal:** Create a fast, user-friendly, and responsive frontend so users can interact with AI-driven budget recommendations.
- **Supported Categories:** Home Interior, Party Planning, and Jewelry.

- **Backend Integration:** Links directly with a FastAPI backend to take user inputs in real time and display structured AI responses.
- **Key Features:** User authentication, personalized session management, multi-category input, and clear recommendation displays.

❖ Activity 4.1: Develop HTML Frontend Interface

- Design a home page with budget input fields and category selectors (Home, Party, Jewelry)
- Implement individual forms for each planner module connected to FastAPI POST routes
- Display AI responses in well-structured, card-like layouts (e.g., product suggestions, item details, prices)

Templates Directory Structure:



MILESTONE 5: Testing & Optimization :

- **Goal:** Ensure PocketSmart AI's recommendation engine is accurate, reliable, and context-aware.

- Optimization: Fine-tuning AI prompts, improving session handling, and ensuring smooth UI/UX across all devices.

❖ **Activity 5.1: Test with Real-World Budget Inputs Across Domains**

- Conduct end-to-end testing using varied budgets and user preferences for:

- Home decor setups from platforms like IKEA, Amazon
- Party packages including food (Swiggy/Zomato), venues (OYO), and decor
- Jewelry suggestions matched with outfit styles, occasions, and image inputs

✓ **Home Page**

Home Page: The main landing page introducing PocketSmart AI's features with links to planner modules and user actions.

PocketSmart

AI-Powered Budget Planning for Everyday Needs

Make smarter financial decisions with personalized budget recommendations for home interiors, parties, and jewelry purchases. Our AI helps you get the most value for your money.

[Get Started](#)[Learn More](#)

Our Smart Budget Planners

Discover how PocketSmart helps you make better financial decisions across different areas of your life



Home Interior Budget Planner

Get personalized recommendations for furniture, lighting, and decor that fit your style preferences and budget constraints. Our AI helps you create a beautiful space without overspending.



Party Budget Planner

Plan your perfect event with smart budget allocations for venue, catering, decorations, and entertainment. Our AI suggests the best ways to create memorable events while staying within your budget.



Jewelry Budget Planner

Find the ideal jewelry pieces for any occasion that match your outfit and budget. Our AI recommends options based on your style preferences, occasion, and available budget.

✓ Testimonials Page

Testimonials Page: Showcases real user reviews and success stories to demonstrate the platform's effectiveness.

What Our Users Say

Real experiences from people who have transformed their financial planning with PocketSmart

"PocketSmart helped me furnish my new apartment without breaking the bank. The recommendations were spot on and I saved nearly 30% of my original budget!"

 **Sarah K.**
Home Owner

"Planning my daughter's birthday party was so much easier with PocketSmart's budget breakdown. The AI suggestions for affordable decorations and catering options were fantastic."

 **Michael R.**
Parent

"The jewelry recommendations perfectly matched my outfit for the wedding. Saved me hours of searching and I received so many compliments on my accessories!"

 **Priya M.**
Fashion Enthusiast

Ready to Optimize Your Budget?

Join PocketSmart today and start making smarter financial decisions across all areas of your life.

[Sign In](#)

[Create Account](#)

✓ Footer Part

✓ Register Page

Register Page: Allows new users to create an account to access personalized budget recommendations.

PocketSmart

AI-Powered Budget Planning

Create Your Account

Username

 Choose a username

Email

 Enter your email

Password

 Create a strong password

Confirm Password

 Confirm your password

Create Account 

Already have an account? [Sign In](#)

Login Page: Authenticates existing users to access their dashboard and saved data.

PocketSmart

AI-Powered Budget Planning

Welcome Back

Username



Enter your username

Password



Enter your password

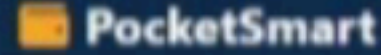
[Forgot Password?](#)

Sign In →

Don't have an account? [Create Account](#)

✓ User Dashboard

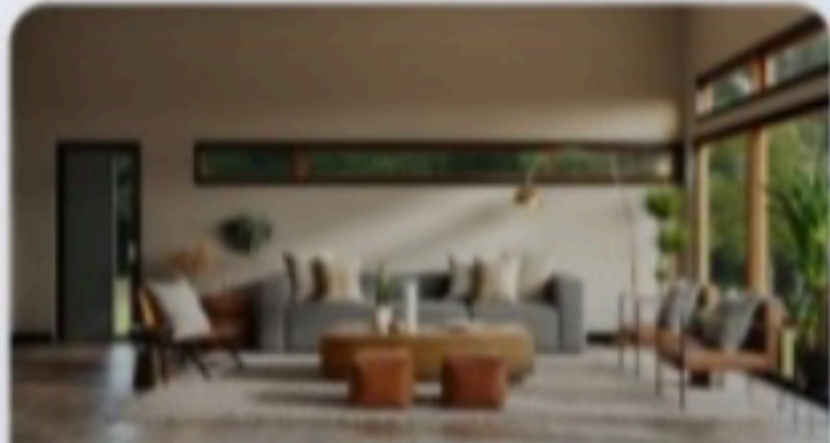
User Dashboard: Displays recent recommendations, saved queries, and personalized insights for the logged-in user.



 Home Planner
  Party Planner
  Jewelry Planner
  History
  Logout

Welcome, sai!

Choose a budget planner to get started with your personalized financial planning experience



 **Home Budget Planner**

Plan your interior design budget efficiently with AI-powered recommendations for furniture, lighting, and more.

Get Started



 **Party Budget Planner**

Plan your perfect event with budget allocations for venue, catering, decorations, and entertainment.

Get Started



 **Jewelry Budget Planner**

Find the ideal jewelry pieces for any occasion that match your outfit and stay within your budget.

Get Started

View All Recommendation History

View All Recommendation History

Recent Activity

 **Home Budget Plan**
Budget: \$5,000 - Created on May 2, 2025

 **Birthday Party Budget Plan**
Budget: \$1,200 for 30 guests - Created on May 5, 2025

 **Jewelry Budget Plan**
Budget: \$800 for Anniversary - Created on April 28, 2025

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Home Interior Budget Planner Page: Lets users input home decor preferences and budget to get tailored suggestions from multiple platforms.

Home Interior Budget Planner

Create a customized budget plan for your dream home interior

Budget Details

Total Budget (\$)

 \$000

Fixtures & Furniture

 Number of Lights/Fixtures

5

 Number of Ceiling Fans

4

 Number of Furniture Pieces

2

 Number of Dining Tables

1

Rooms to Include

☒  Living Room

☒  Kitchen

☐  Bedroom

Additional Information

Special Requirements or Preferences

Any specific requirements or preferences...

 **Generate Recommendations**

Home Interior Recommendations Page: Displays AI-generated product suggestions for home decor based on the user's budget and style preferences.

[Generate Recommendations](#)

Your Personalized Budget Plan

Budget Summary

Total Budget: \$5000.00

Remaining Budget: \$500.00

Lighting

Allocation: \$1500.00

Item	Description	Price	Quantity	Shopping Links
LED Bulb (Warm White)	Energy-efficient LED bulbs for general lighting.	\$100.00	5	Amazon Flipkart Ikea Myntra Ajio

Ceiling_fans

Allocation: \$2000.00

Item	Description	Price	Quantity	Shopping Links
Havells Ceiling Fan	Basic, functional ceiling fan.	\$500.00	4	Amazon Flipkart Ikea Myntra Ajio

Furniture

Allocation: \$1000.00

Item	Description	Price	Quantity	Shopping Links
Plastic Chair	Stackable plastic chairs for kitchen or living room.	\$250.00	2	Amazon Flipkart Ikea Myntra Ajio
Small Wooden Table	Simple wooden table for dining or side table.	\$500.00	1	Amazon Flipkart Ikea Myntra Ajio

Additional Suggestions

- Consider purchasing used furniture for further cost savings.
- Look for sales and discounts on online marketplaces.
- Prioritize essential items and postpone non-essential purchases.

Party Budget Planner Page: Helps users plan events by suggesting venues, food, and decorations based on budget and guest count

PocketSmart

Home PlannerParty PlannerJewelry PlannerHistoryLogout

Party Budget Planner

Plan your perfect event with AI-powered budget recommendations

\$ Basic Information

Total Budget (₹)

₹ 5000

Number of Guests

3

🏠 Event Details

Party Type

📅 Wedding

Venue Type

📍 Home

📋 Party Needs

☒ 🍽️ Catering

☐ 🎨 Decoration

☒ 🎵 Entertainment

💬 Additional Requirements

Special requests, themes, dietary restrictions, etc.

Generate Budget Plan

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Party Budget Recommendations Page: Shows curated party planning options, including food, venues, and decorations, matched to the provided budget.

Your Party Budget Plan

[Print/Save](#)

Budget: ₹5000.00

Category	Item	Amount
Venue	Home	₹0.00
	Utilizing the home as the venue.	
	Shop on: Google Booking MakeMyTrip OyoRooms Nabroker	
Catering	Home-cooked meal	₹2000.00
	Simple, home-cooked meal for 3 people.	
	Shop on: Swiggy Zomato	
Entertainment	Streaming service subscription	₹500.00
	One-time purchase of a movie or renting a few movies on a platform like Netflix, Amazon Prime Video, or Hotstar.	
	Shop on: Amazon Flipkart Bookmyshow	
	Board games/cards	₹500.00
	Purchase of a new board game or card game.	
	Shop on: Amazon Flipkart Bookmyshow	
	Music playlist	₹0.00
	Creating a celebratory playlist on Spotify or YouTube Music.	
	Shop on: Amazon Flipkart Bookmyshow	
	Small gift for couple	₹1000.00
	A small token of appreciation for the couple.	
	Shop on: Amazon Flipkart Bookmyshow	
Contingency	Unexpected expenses	₹1000.00
	Buffer for unforeseen costs.	
	Shop on: Amazon Flipkart Google	
Total Budget		₹5000.00
Allocated		₹5000.00
Remaining		₹0.00
Venue Suggestions	Home	₹0.00
	Type: Residential	
	Location: undefined	
	undefined	
	Website View on Map	
Additional Suggestions	- Consider making the meal a potluck style if comfortable with guests to reduce catering costs. - Look for discounts or offers on streaming services or board games. - Homemade decorations can be a cost effective alternative if you decide to add some.	

Jewelry Budget Planner Page: Provides jewelry recommendations aligned with occasion and outfit style, using text and optional image inputs.

PocketSmart

DashboardHome BudgetParty BudgetJewelry Budget

Logout

Jewelry Budget Planner

Get AI-powered jewelry recommendations within your budget for any occasion

\$ Budget Details

Total Budget (\$)

\$5000

📅 Occasion & Preferences

📅 Occasion

Birthday

💖 Style Preferences

Describe your style preferences, materials, colors, etc.

👕 Upload Outfit Image

Remove Image

Get Recommendations

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Jewelry Budget Recommendations Page: Presents personalized jewelry options tailored to the occasion, outfit, and budget, using Gemini-generated insights.

 Get Recommendations

Your Personalized Jewelry Recommendations

 Budget Summary

Total Budget: \$5000.00

Remaining Budget: \$800.00

 Outfit Analysis

Colors: blue, white

Style: casual

Formality: informal

 Jewelry Recommendations

 bracelet

Description: A simple, braided leather bracelet with metal accents. This complements the casual style of the outfit without being overly flashy.

Style: casual

Shop For This:

 Amazon
  Flipkart
  Bluestone
  Tanishq
  Carallane
  Maloma
  Meesho

 ring

Description: A silver or dark grey metal ring with a minimalist design. Avoid anything too large or ostentatious to maintain the casual feel.

Style: minimalist

Shop For This:

 Amazon
  Flipkart
  Bluestone
  Tanishq
  Carallane
  Maloma
  Meesho

 watch

Description: A classic, simple watch with a leather or metal band. A darker band would complement the shirt's colors.

Style: classic

Shop For This:

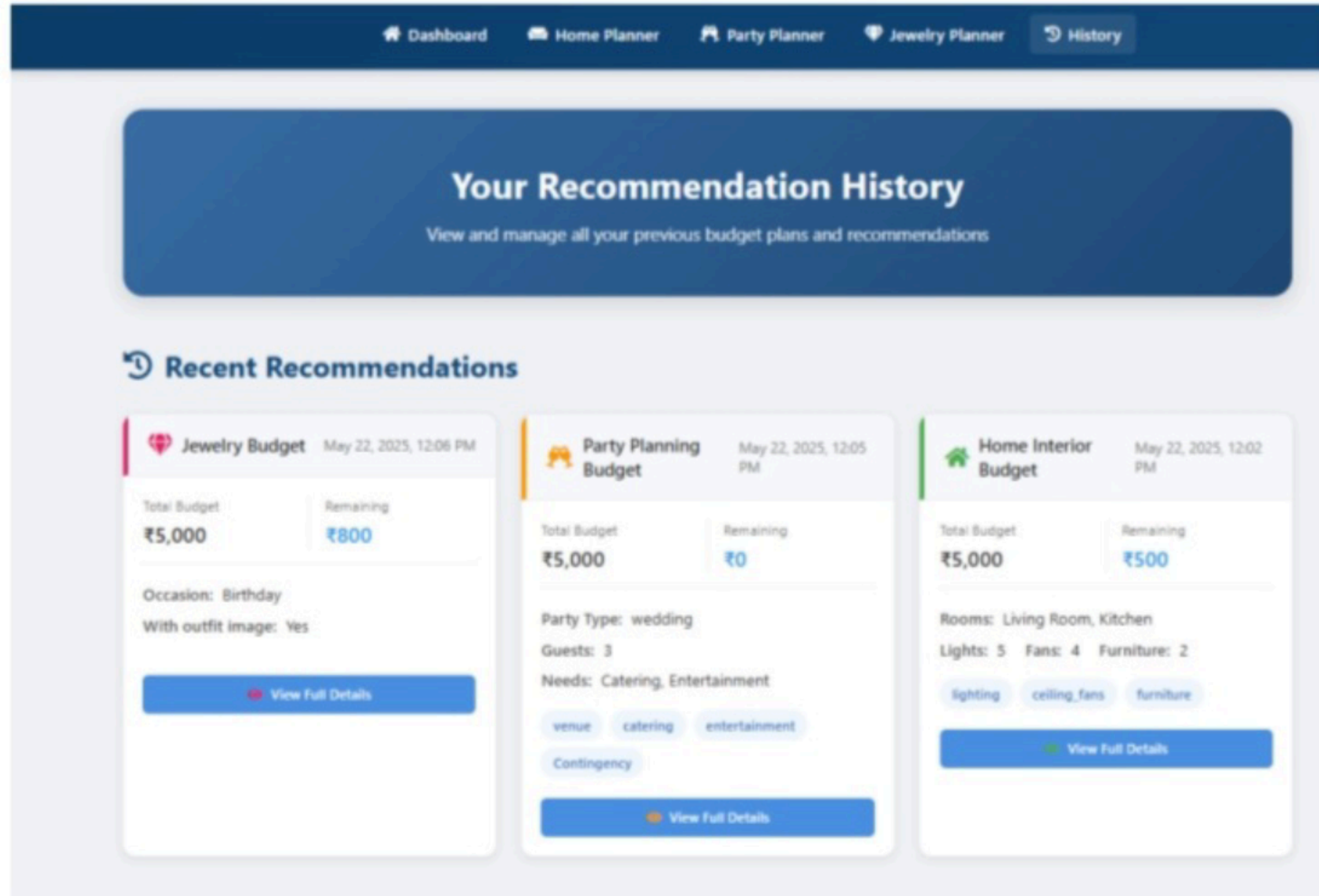
 Amazon
  Flipkart
  Bluestone
  Tanishq
  Carallane
  Maloma
  Meesho

 Styling Tips

- Keep the jewelry minimal to match the casual style of the outfit.
- Consider the watch as a statement piece, choosing a design that reflects personal style.
- Ensure the metal tones of the ring and bracelet (if metal accents are chosen) complement each other.

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Recommendation History Page: Displays a log of the user's past recommendation queries and results, allowing easy review and reuse of previous plans.



The screenshot shows the 'Your Recommendation History' page. At the top, a navigation bar includes links for Dashboard, Home Planner, Party Planner, Jewelry Planner, and History. The main heading is 'Your Recommendation History' with a subtitle 'View and manage all your previous budget plans and recommendations'. Below this, a section titled 'Recent Recommendations' displays three cards:

- Jewelry Budget** (May 22, 2025, 12:06 PM): Total Budget ₹5,000, Remaining ₹800. Occasion: Birthday. With outfit image: Yes. View Full Details.
- Party Planning Budget** (May 22, 2025, 12:05 PM): Total Budget ₹5,000, Remaining ₹0. Party Type: wedding. Guests: 3. Needs: Catering, Entertainment. View Full Details.
- Home Interior Budget** (May 22, 2025, 12:02 PM): Total Budget ₹5,000, Remaining ₹500. Rooms: Living Room, Kitchen. Lights: 5, Fans: 4, Furniture: 2. View Full Details.

Conclusion :

PocketSmart AI is an intelligent, AI-driven budget and lifestyle recommendation assistant that seamlessly integrates FastAPI, Gemini 1.5 Flash Pro, and a Jinja2-powered frontend. By processing user budgets, preferences, and images, the platform delivers context-aware suggestions across domains like home interiors, party planning, and jewelry via trusted services such as Amazon, Flipkart, IKEA, Swiggy, Zomato, and OYO. Designed with a focus on scalability and user experience, the system automates decision-making and streamlines everyday planning into a fast, personalized, and budget-friendly experience.